



Food For Thought: Sowing The Seeds of Food Inflation

July 15, 2026

Ongoing tensions in the Middle East have raised the prospect of indirect effects on inflation - particularly in relation to food prices. Indeed, much has been said about the rise in energy prices and its spill over into other commodities such as fertiliser, CO₂, and other key ingredients that sit within the agricultural pipeline. To be sure, both the size and duration of the shock matter for food prices.

In this note, we do four things.

- **First**, we highlight what's happened to key commodity prices in the wake of the Iran shock.
- **Second**, we set out a framework, estimating the potential impact of commodity price moves on food inflation for both the euro area and the UK.
- **Third**, we stress test our estimates by looking at the duration of the shock and whether pass-through of cost increases change when the shock is deemed 'transitory' or 'persistent'.
- **Finally**, we highlight some of the broader implications for inflation and where risks lie around our estimates.

Our analysis points to renewed upside risks to food prices stemming from recent oil and fertiliser market developments. The effects appear larger in the UK than in the Euro area.

Importantly, we find that pass-through isn't immediate, and in fact, can build over a long period of time given the mapping of commodity prices into food retail prices. What also matters is whether shocks prove persistent or transitory. Notwithstanding the MoU agreed to by the US and Iran, we estimate that the net effects from oil and urea prices will lead to modest inflationary pressure in the quarters ahead – something we will be watching for closely. That said, considerable uncertainty surrounds both the scale and timing of these effects, with a range of other supply-side factors also likely to influence outcomes. We will be monitoring market developments closely and will calibrate our projections accordingly.

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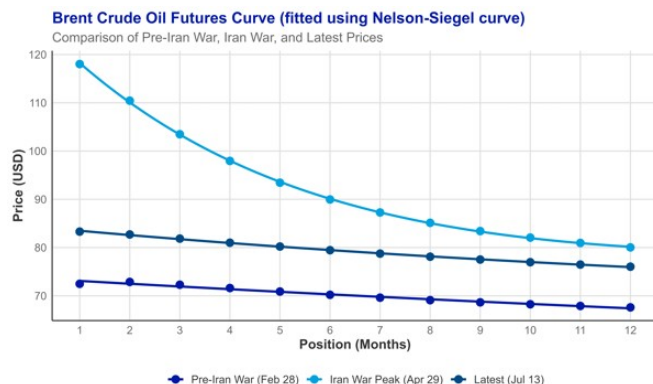


A recap on recent price movements

Recent geopolitical developments have been associated with large swings in fertiliser and oil prices. Oil prices initially rose as markets priced supply-disruption concerns. Subsequently, however, price rises retraced meaningfully as these concerns eased with continuing peace deal negotiations. To add context to these movements, we compare spot and futures prices before the episode, after the initial Iran shock, and using the latest available market data. For the futures curves, we fit a Nielsen-Siegel curve to trace out the shape of the curve across contract positions.

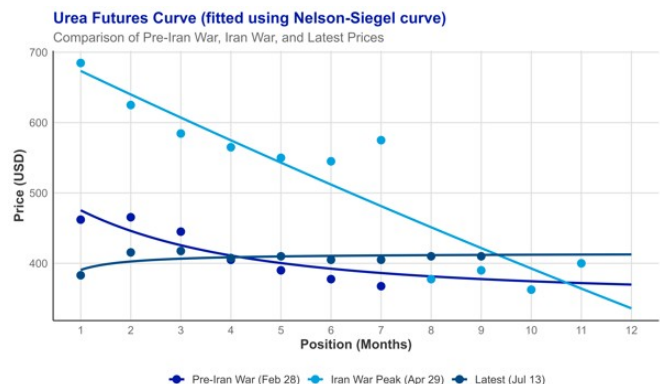
For fertiliser prices, we use the World Bank urea price series as a proxy for global nitrogen fertiliser costs. Urea is one of the most widely traded nitrogen fertilisers and is closely linked to the production process for agricultural inputs. We note, however, that this captures only one important fertiliser-cost channel rather than the complete range of farm input costs.

Figure 1: Brent futures pricing downward sloping...



Source : Deutsche Bank, Bloomberg Finance LP

Figure 2: ...same as urea futures pricing



Source : Deutsche Bank, Bloomberg Finance LP

Before the Iran War, the spot price for Brent crude oil was trading at about USD 72.50/barrel. This then spiked to USD 118/barrel during the peak of the Iran/US conflict, before settling at around USD 83/barrel at current market prices ([figure 1](#)). Interestingly, the futures curve is currently flatter than the steeper curve observed at the start of the war, which could imply a more gradual decline in prices in the next few months. The urea spot price also increased sharply at the beginning of the Iran war. In contrast, these prices have settled at a much lower level ([figure 2](#)).

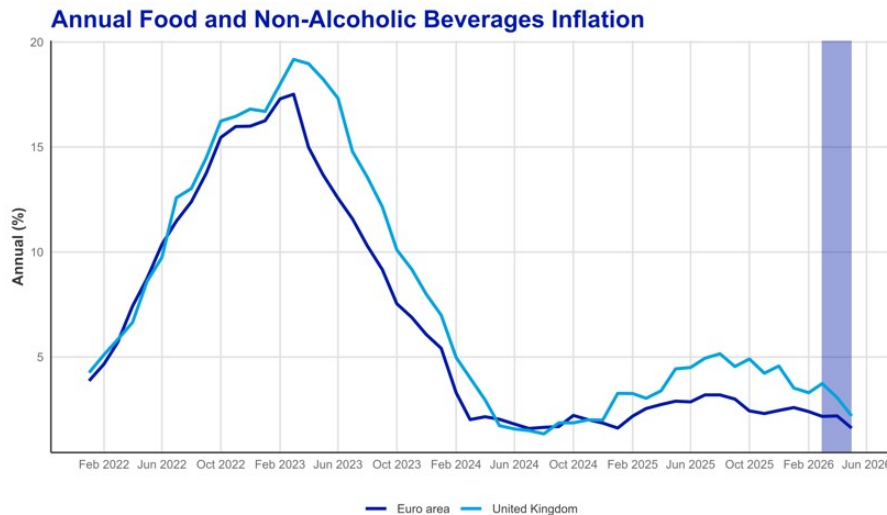
Perhaps unsurprisingly, in terms of recent food price developments, we have yet to see any tangible impact on retail prices.

Euro area food inflation has softened more than expected this year, declining from 2.5% y-o-y in Dec-25 to 1.6% in Jun-26 based on the flash estimate. This marks the lowest reading for food HICP since mid-2021, with the bulk of the adjustment occurring over the past two months. Focusing on food stuff, the largest disinflationary impulses have come from dairy products, sugar and sweets, and coffee. In 2025, these categories were among the main drivers of food inflation, accounting for roughly 35% of annual food inflation despite representing only 18% of the HICP food basket. In recent months, their contribution has fallen to just over 20%. Meat prices have also eased somewhat from their 2025 peak, although they remain the largest contributor to food inflation.



In the UK, we see a similar story. Heavy discounting has led to a drop-off in food prices. Food and non-alcoholic beverages CPI slowed to 2.2% y-o-y – reaching its lowest rate in 17 months. The slowdown in food prices can be seen across both processed and unprocessed food. The latest PPI data for food points to further disinflation in the pipeline, with producer prices across food, beverages and tobacco slowing to 1.1% y-o-y in May-26.

Figure 3: Food (and non-alcoholic beverages) inflation in the EA and the UK has been on a downtrend this year



Source : Deutsche Bank, Macrobond

How does energy and fertiliser costs pass-through into food inflation? There are two channels in which oil and fertiliser prices work their way into CPI.

Channel 1: the direct energy cost. Oil enters food at pretty much every node in the food chain. For example, manufacturers need diesel for tractors and irrigation pumps, marine freight and trucking, processing heat, packaging (plastics) and refrigeration. Energy-related inputs account for around 5–10% of farm production costs, although the share can be significantly higher in energy-intensive sectors such as horticulture and poultry.

Channel 2: fertiliser cost. Urea is essentially manufactured from natural gas. The higher gas prices go, the higher fertiliser prices go too. The response can effectively be summarised into two segments: an immediate margin squeeze, then a supply response as farmers cut application rates or switch acreage to lower Nitrogen crops (soybeans over maize), which lifts prices with a lag of a full crop cycle.

The transmission from input costs to retail prices, as a result, is heavily staggered. The upstream leg is fast: ammonia and urea reprice off gas within weeks. Put simply, an energy shock is visible in fertiliser markets almost immediately. The middle leg is slow and seasonal — application decisions are made pre-season, so a nitrogen cut driven by a spring price spike doesn't show up until (as early) as the following harvest. Ultimately, the final leg is both lagged and heavily damped.



Estimating pass-through to food prices

There are several ways to estimate the effect of oil and fertilizer price shocks to food inflation. In this case, we opt for a simple approach that allows the "data to speak": the local projection method ([Jorda, 2005](#)). Essentially what we do is trace out the impact of a shock over time, in this case oil and fertilizer price shocks, by running a separate regression for each forecast horizon (1-month ahead, 2-months ahead and so on...).

A key challenge is identifying the commodity-price shock itself. Oil and fertiliser prices move for many different reasons, including global demand, exchange rates, trade conditions, inflationary pressures and supply disruptions. To reduce the risk of conflating commodity-specific surprises with broader macroeconomic developments, we use a Romer and Romer-style approach ([Romer & Romer, 2004](#)).

The above approach involves regressing the shocks of interest on a set of potentially confounding variables, including the USD effective exchange rate, monthly dummy variables to capture seasonality, lags of the shocks of interest to account for persistence, global trade volumes and global CPI inflation. We then extract the residuals from this regression. These residuals could then be interpreted as movements in oil and urea prices that are not explained by our macroeconomic controls.

The shocks of interest that we use are the month-on-month percentage changes in Brent crude oil spot prices and the World Bank urea commodity price series.

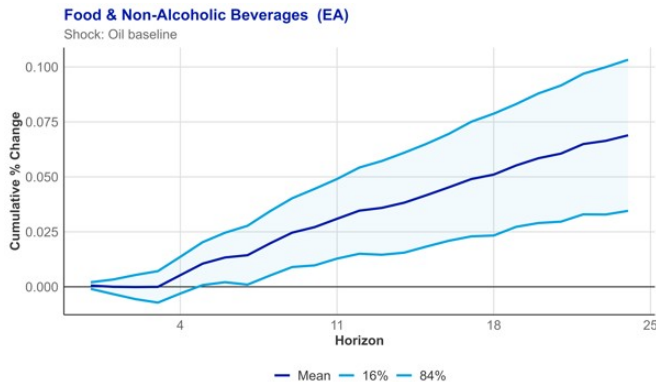
For the local projections, we use the HICP Food & Non-alcoholic beverages series for the Euro area and the UK Consumer Price Index Food and Non-alcoholic beverages. The regressions include the shock of interest (oil & urea), and three lags of the price variables. We then estimate the cumulative percent impact on the price level from a percentage point surprise in both oil and urea costs.

Baseline results

The baseline estimates suggest that oil and fertiliser price surprises have a measurable effect on food prices. The estimated response varies across the two economies and commodities, with the UK showing larger pass-through than the euro area in our specifications. Our estimates, however, become relatively more uncertain the further out we go as shown by the widening error bands.

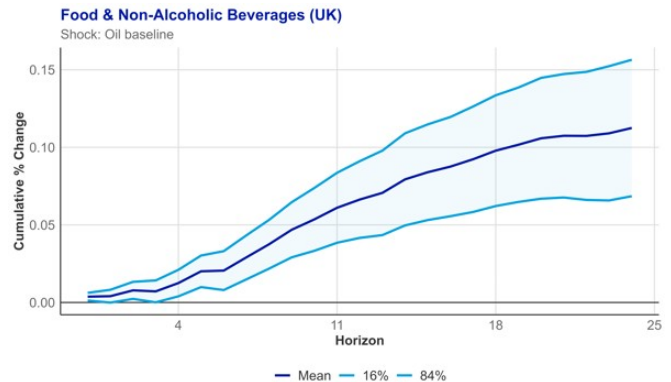


Figure 4: A 1pp shock in oil price leads to nearly a 7bps increase in food HICP (EA)...



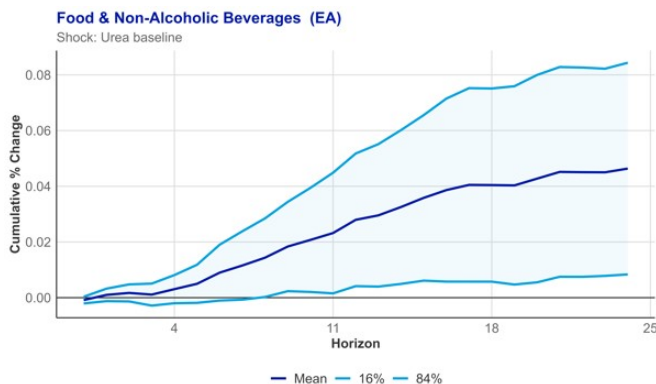
Source : Deutsche Bank

Figure 5:...and just over 10bps in the UK (CPI)



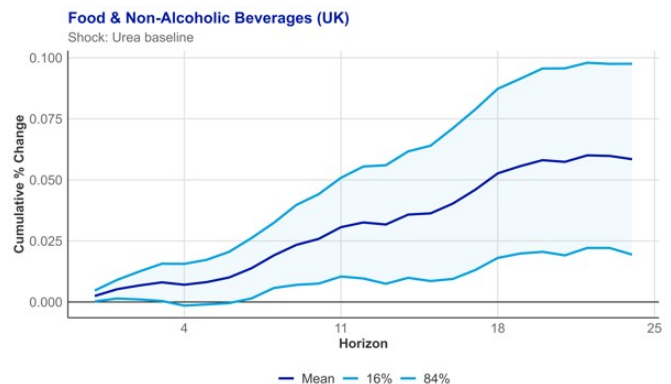
Source : Deutsche Bank

Figure 6: A 1pp shock to urea prices results in just over 4bps to EA HICP (food & non-alcoholic beverages)...



Source : Deutsche Bank

Figure 7:...and close to 6bps for UK CPI (food and non-alcoholic beverages)



Source : Deutsche Bank

In further detail, [figures 4-7](#) present the impact of a 1 percentage point increase in monthly oil and urea price growth on the cumulative food & non-alcoholic beverages price level. Looking first at fertiliser price shocks, we find that the peak impact on UK food prices occurs after around 20 months with a 0.07% cumulative impact on the food price level ([figure 7](#)). In the Euro area, the peak impact is slightly lower at around 0.05% and also occurs after roughly 20 months ([figure 6](#)). A broadly similar pattern is visible for oil price shocks. The UK experiences a larger cumulative increase of 0.1% in food prices while the Euro area sees an impact of about 0.07%.

There are a few possible explanations for the slightly larger estimated effects in the UK relative to the Euro area. One possible explanation that may be driving the difference in pass-through between the Euro area and the UK is the higher imported share in the UK food basket. This would make importers and producers more exposed to externally driven input cost pressures.

A second explanation may come from sterling weakness during global energy shocks.

A third explanation may be in fact that many UK farming sectors operate on narrow margins. When energy, fertiliser or feed costs rise sharply, farmers have less ability

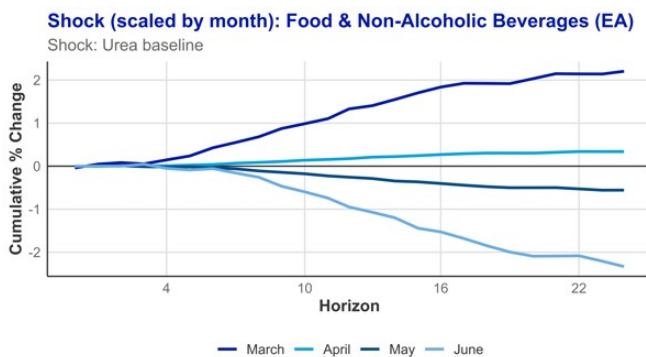


to absorb the shock and are more likely to pass it on through the supply chain. Defra's Farm Business Survey consistently shows that profitability is highly sensitive to movements in input costs. UK agriculture is also relatively intensive, particularly in cereals, oilseeds and grassland systems.

The impact of the Iran war

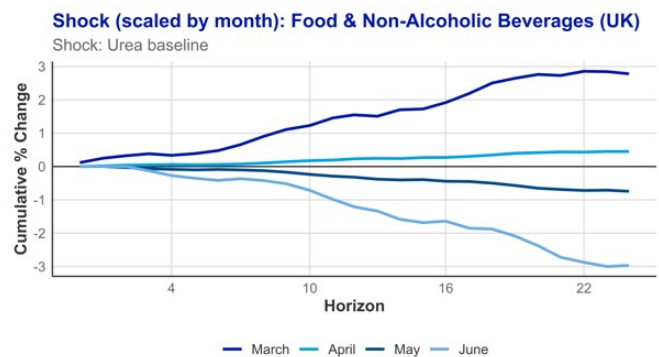
So far, the results have shown a 1 percentage point increase in fertiliser and oil price inflation. However, events such as the Iran war have led to big swings in both urea and oil prices – and in both directions. In order to better gauge these impacts, we scale the shocks in the baseline results to the actual movements observed from March to June. The figures below show the estimated cumulative food price pass-through implied by each monthly commodity price move.

Figure 8: Estimated urea shock pass-through for each month since the conflict began in the EA...



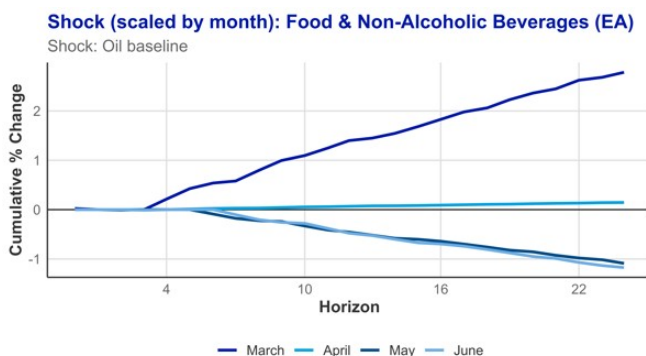
Source : Deutsche Bank

Figure 9:...and the UK



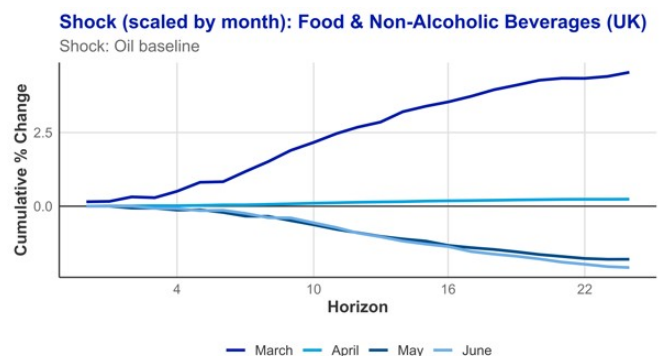
Source : Deutsche Bank

Figure 10: Scaled oil shock between March and June for the EA...



Source : Deutsche Bank

Figure 11: ...and the UK



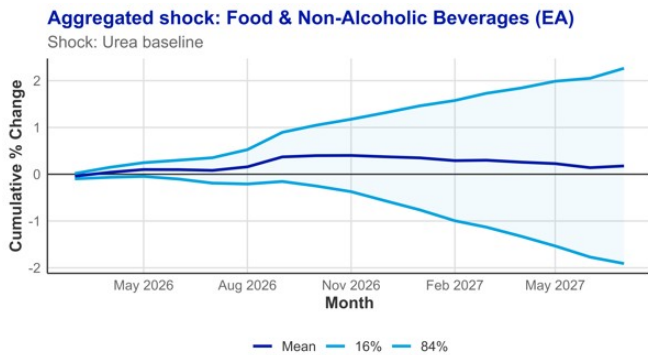
Source : Deutsche Bank

To be sure, the March shock impact alone would have had a big impact on food prices in the Euro area and the UK. For both oil and urea, it was an upside surprise of over 40 percentage points (over 3 standard deviations). However, over the course of the past few months, there have been significant reversals in both oil and urea prices. We attempt to capture these offsetting shocks and obtain the overall impact since March. How do we do this? We do this by aggregating the impacts of the



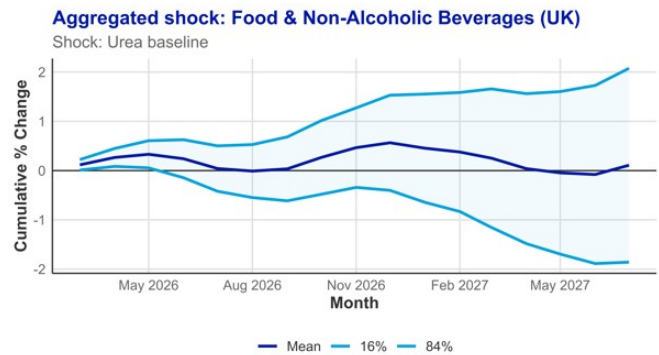
monthly urea and oil price shocks – i.e. we aggregate the shocks from oil and urea via superposition of the estimated local projections.

Figure 12: Estimated aggregate urea shock on food HICP in the EA...



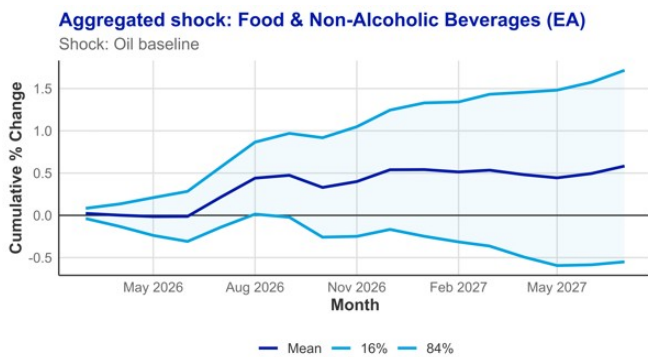
Source : Deutsche Bank

Figure 13: ...and on UK food CPI



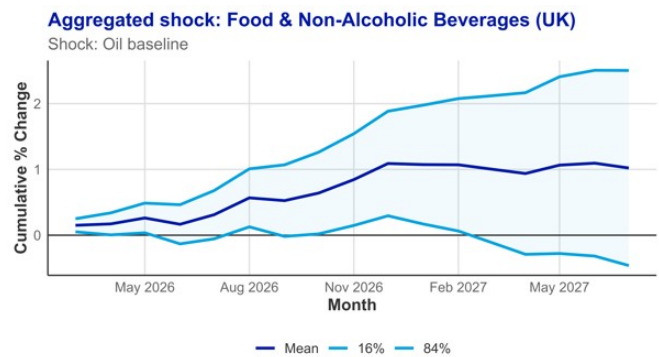
Source : Deutsche Bank

Figure 14: Estimated aggregated oil shock on food HICP in the EA...



Source : Deutsche Bank

Figure 15: ...and on UK food CPI



Source : Deutsche Bank

We trace out these shocks and find some interesting implications for the overall impact. For the UK, we see a cumulative impact of around 1% on food prices from the oil shocks a year from March and 0.3% from a urea shock peaking also in a year's time from March, i.e. totalling nearly 130bps (though the effect starts to unwind four quarters out). For the Euro area, the oil shocks lead to about a 0.5% cumulative increase in a year from March and about a 0.3% impact from urea shocks in a year from March. Overall, we observe similar patterns where the UK experiences a slightly stronger impact from the recent shocks compared to the Euro area.

Stress testing our results: persistent vs transitory shocks

A simple scaling of the shocks is unlikely to fully reflect firms' pricing behaviour. In practice, firms may respond differently to the size and duration of the shock. A large but temporary increase in oil or fertiliser prices may generate less pass-through than a smaller but more sustained rise.

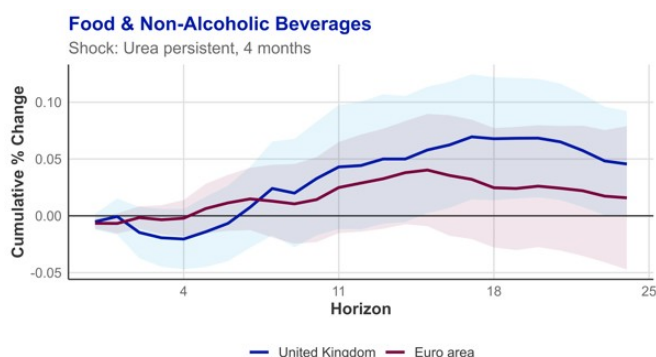
To capture this distinction, we split our oil and fertiliser shocks between persistent



and transitory. We classify a shock as persistent if the price remains at least half as large after four months. If the price falls by more than half over that period, we treat it as transitory. We only apply this classification to sizeable shocks, defined as moves larger than one standard deviation in the relevant shock series. This gives us a simple way to separate temporary price spikes from more sustained input-cost pressures.

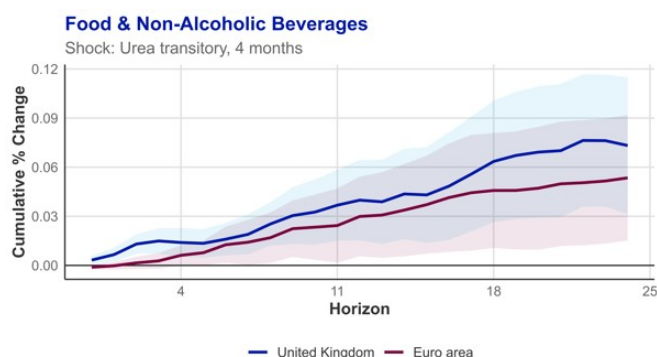
We then run local projections again on the persistent and short-lived shocks and we obtain the following results:

Figure 16: Persistent urea price shock...



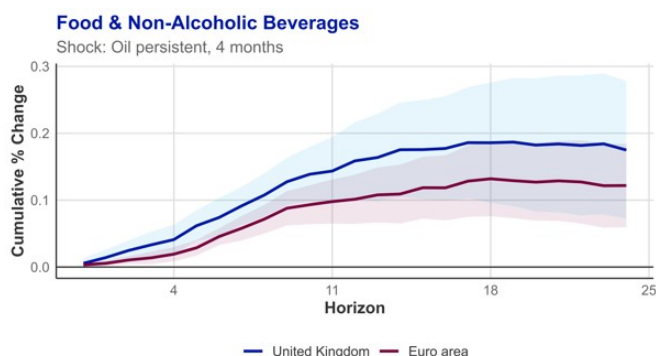
Source : Deutsche Bank

Figure 17: ...vs transitory urea price shock



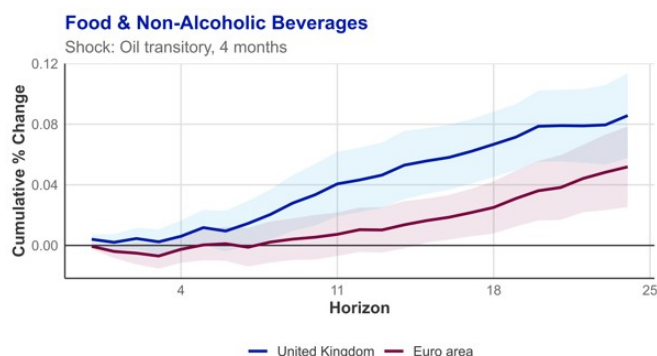
Source : Deutsche Bank

Figure 18: Persistent oil price shock...



Source : Deutsche Bank

Figure 19: ...vs transitory oil price shock



Source : Deutsche Bank

The results suggest that persistence matters, particularly for oil shocks ([figures 18 and 19](#)). When the shock is persistent, we find a stronger peak impact for both the Euro area and the United Kingdom for a given 1 percentage point surprise compared to a short-lived shock. For urea, the pattern is clearer in the UK, where persistent shocks are associated with a larger food price pass-through. In the Euro area, however, the difference between persistent and short-lived shocks is less clear. Overall, the evidence points to a stronger role for shock persistence in the UK, while the Euro area results are more mixed.

While it is too early to definitely classify the latest energy price shocks, the above results highlight an important distinction. If recent recent rises in oil and fertiliser prices prove short-lived and unwind over the coming months, then pass-through to



food prices may be more limited. However, if prices remain elevated, history suggests we could expect a larger and more sustained impact on food inflation.

Based on the shocks observed so far, our classification leans towards the recent developments being transitory in nature. That said, the classification is far from certain and will depend on how oil and fertiliser prices evolve through July and the months ahead.

Conclusions

What to make from the above analyses? We highlight several key findings.

First, the impact of a fertiliser and oil price shock and its pass-through to food price levels tends to be stronger in the United Kingdom than in the Euro area.

Second, after scaling and aggregating the fertiliser and oil price shocks from March up to June, we find that in a year's time from March, the cumulative impact on the UK food price level could be about 1.3% while for the Euro area it would be around 0.8% instead – implying a 0.1-0.15pp increase in headline HICP/CPI.

Third, we also recognise the limitations of simply scaling these shocks. The size and duration of the shock can result in very different implications for food price pass-through. The persistence of a shock can have a much larger impact, especially in the UK. It's too early to say where the current energy shock will fall into. Recent price action suggests a further wave of energy pressures on the back of escalating Middle East tensions. We will be monitoring this closely. The longer commodity prices remain elevated, the more they feed through into food prices.

Lastly, the confidence bands around our estimates suggest wide variation in pass-through. Given the swings in price rises across oil and urea, the staggered pass-through of cost rises into the food supply chain may be more varied. While our estimates provide a measure of the average conditional pass-through, other factors will also impact pass-through. For example, warmer global temperatures, less rainfall, and changing labour costs alongside ongoing geopolitical tensions may also shape pricing strategies.

To be sure, despite the pull back in oil and urea prices from their peak, the net effects still point to a modest jump in food (and non-alcoholic beverages) prices. Furthermore, the recent jump back in price momentum will do little to assuage concerns of indirect effects brewing within inflation. We will reflect some of our findings over the coming week(s) as we update our inflation projections.



Appendix 1

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